

LAB 1 — COMPLETE REQUIREMENTS TABLE

Problem Statement: Blood Bank Inventory & Emergency Donor Matcher

Exactly 5 Functional Requirements (FR-001 to FR-005) and 2 Non-Functional Requirements (NFR-001 and NFR-002), using the required fields from the handout.

Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow the emergency requester to create a blood request by entering the required blood group and number of units.	High	When valid blood group and unit details are entered, the system creates the request and displays a request ID.	Required to request blood during an emergency.
FR-002	Functional	The system shall allow the blood bank manager to add and update blood stock details.	High	When the manager adds or updates a blood unit, the new quantity is displayed in the inventory.	Keeps blood stock information updated.
FR-003	Functional	The system shall check the available blood stock when an emergency request is submitted.	High	For a submitted request, the system displays whether the required blood group and quantity are available.	Helps quickly identify available blood.
FR-004	Functional	The system shall allow the blood bank manager to issue available blood units for an approved request.	High	When blood is issued, the issued quantity is deducted from the inventory.	Maintains correct blood stock after issue.
FR-005	Functional	The system shall search for suitable donors when the required blood is not available in the inventory.	High	When stock is insufficient, the system displays available donors matching the required blood group.	Helps find blood during emergencies.

Non-Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Nonfunctional	The system shall display blood inventory and donor matching results within 5 seconds of a valid request.	High	A timing test shows that the result is displayed within 5 seconds.	Quick response is important during emergencies.
NFR-002	Nonfunctional	The system shall allow only authorized users to access blood bank and donor information.	High	An unauthorized user attempting to access restricted information is denied access.	Protects sensitive healthcare and donor information.